“Futuristic Radar Object Detection With Camera”

A DESIGN PROJECT-1 (5CS1990 / 5IT1990) REPORT

Submitted by

NAME	REGISTRATION NO.
CHITTAGARI ARUN KUMAR REDDY	2023BCSE07AED195
SHAIK MOHAMMED MAHABOOB BASHA	2023BCSE07AED211
GONAPATI VISHADHAR REDDY	2023BCSE07AED192
B.C.V.GANESH	2023BCSE07AED179

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Dr. Virendra Kumar Shrivastava

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ALLIANCE UNIVERSITY

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ALLIANCE UNIVERSITY, BENGALURU

DEPARTMENT OF CSE (AIML)

CERTIFICATE

This is to certify that the Design Project -1 work entitled **"Futuristic Radar Object Detection With Camera"** is the bonafide work done by CHITTAGARI ARUN KUMAR REDDY(2023BCSE07AED195),SHAIKMOHAMMEDMAHABOOBBASHA(2023BCSE07AED211),GONAPATIVISHADHARREDDY(2023BCSE07AED192),B.C.V.GANESH (2023BCSE07AED179), submitted in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in DEPARTMENT CSE-AIML during the year 2025-2026, IV-semester.

Under Supervisor of

Professor. Virendra Kumar Shrivastava

Dept. of CSE(AIML)

ALLIANCE UNIVERSITY

ALLIANCE SCHOOL OF ADVANCED COMPUTING

DECLARATION

This is to declare that the report titled "**Futuristic Radar Object Detection With Camera"** has been made for the partial fulfilment of the Programme - Bachelor of Technology in **Computer Science and Engineering** , under the guidance of **Dr.Virendra Kumar Shrivastava.** We confirm that this report truly represents my work undertaken as a part of **my Design Project -1** work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the Mentor.

NAME	REGISTRATION NO.	SIGNATURE
CHITTAGARI ARUN KUMAR REDDY	2023BCSE07AED195	
SHAIK MOHAMMED MAHABOOB BASHA	2023BCSE07AED211	
GONAPATI VISHADHAR REDDY	2023BCSE07AED192	
B.C.V.GANESH	2023BCSE07AED179	

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Student names:

1.)CHITTAGARI ARUN KUMAR REDDY

2.)SHAIK MOHAMMED MAHABOOB BASHA

3.)B.C.V.GANESH

4.)GONAPATI VISHADHAR REDDY

ABSTRACT

This project introduces **Neon Nexus v7.4**, an innovative awareness system designed to provide real-time, multi-modal feedback about a user's surroundings, primarily addressing the challenge of navigation in low-visibility or for the visually impaired. The system fuses two distinct technologies within a single, browser-based application: AI-powered Object Recognition (using a TensorFlow.js COCO-SSD model) to identify objects by type (e.g., "chair," "person"), and a Physical Arduino Radar (using an ultrasonic sensor) to provide precise distance measurements in centimeters via the Web Serial API. The key innovation is the Intelligent Voice Feedback Logic. Unlike the way other systems can overload users with information, Neon Nexus implements a more nuanced human-centered systems design in that the system announces after detecting new objects, but only speaks distance after the distance reading has been both presented and stable for 3 seconds. This prevents over-hearing from the rapid changes in the sensor data, and system has effects of human-centered ecologies. The system successfully integrates and captures computer vision, microcontroller-data, and web tech, to be robust with low-latency, real-time, and context-sensitive, and that is the basis for their notions for accessibly designed-edge-AI.

LIST OF ABBREVIATION

Abbreviation	Full Form	Description
AI	Artificial Intelligence	The core technology enabling the system to understand and classify objects.
API	Application Programming Interface	A set of rules defining how software components should interact (e.g., Web Serial API).
Baud	Baud Rate	The unit of signaling rate used to measure the speed of data transmission in serial communication.
CM / cm	Centimeter	The metric unit used to measure and report the distance from the sensor.
COCO-SSD	Common Objects in Context - Single Shot Detector	The specific, lightweight deep learning model used for real-time object detection in the browser.
CPU	Central Processing Unit	The primary component that executes instructions (used for model execution in the system).
CSS	Cascading Style Sheets	The language used to design the visual presentation of the web interface.
HC-SR04	Ultrasonic Sensor	The specific model of Ultrasonic Distance Sensor used in the Arduino radar module.
HTML	HyperText Markup Language	The standard markup language for creating the structure of the web interface.

Abbreviation	Full Form	Description
I/O	Input/Output	The communication between an information processing system (like Arduino) and the outside world.
JS	JavaScript	The high-level programming language used for all front-end logic, AI execution, and sensor communication.
LED	Light-Emitting Diode	A semiconductor light source (often used as an indicator light in hardware).
OS	Operating System	The software that manages computer hardware and software resources.
TTS	Text-to-Speech	The technology used to convert written text (object names, distances) into spoken words for auditory feedback.
UI/UX	User Interface / User Experience	The design elements focusing on the clarity and efficiency of the application's presentation and interaction.
USB	Universal Serial Bus	The standard connection used to power the Arduino and establish the serial data link.

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Introduction

Understanding your surroundings is essential for safe and effective mobility. While most individuals use their vision as a primary sense to orient themselves in their environment, individuals who have vision impairments are left to rely on their other senses to effectively mobility in the environment. Our reliance on a single sensory modality to navigate in low visibility, high velocity or complex crowd and environment situations is clearly limited in their nature of considering limited movement. Existing sensing systems for detecting physical obstacles that are commonly utilized, are either complex distance systems or simplistic distance simple sensors that have average feedback where the sensors can give us collective feedback, conflicting feedback or incomplete feedback. We will examine a basic need or basic level of sensory system which is an integrated or intelligent mobile sensing environmental awareness system- through the design and integration of the Neon Nexus v7.4 system made by company Neon Visual Inc. Neon Nexus is not simply a simple functionality based sensory device, rather a layer of artificial intelligence is embedded enough reliable physical sensing hardware. Essentially, Neon Nexus is a real-time visual object recognition system in real-time using a webcam/ipod. Then provides estimated distance using built in sonar or radar controlled by an Arduino. The main- and innovative design is not only visual hazard detection and stigma of physical obstacle sensing, but recently developed and what we are establishing is citing low tech component of integrating Human-Centric Feedback Logic through Text-to-Speech (TTS), which factors out interpreting verbal analog to human processing.This logic is developed as a way to convey transitory or stable information in a consistent manner. Thus, the user will have immediate context (what an object is) and factual numerical information (how far away from the user it is) without being overwhelmed by a barrage of auditory updates that could be confusing. The rest of the report outlines the design methodology, technical construction, and new feedback logic developed to create an accessible, fused, and intelligent sensing solution that increases situational awareness in those who rely on audio information.

Literature Review

Paper 1: Real-Time Object Detection using YOLOv4 and Edge Devices [6]

Abstract: Object detection continues to be one of the primary and most challenging tasks of computer vision and image understanding operations. Considerable advances have been made in object detection via improved object representation and deep neural network models. This paper examines more closely how object detection has developed in the age of deep learning over the last few years.

Accomplishments and Limitations: High power consumption, limited performance on low end devices

Scope: To provide high detection accuracy with optimized inference time.

Paper 2: Avoiding Obstacles for Visually Impaired using Ultrasonic Sensors

Abstract: We categorize these types into three groups anchored-unit, anchor-free, and motor-anchored sensors. Each of these approaches is different based on how they find objects in the image. We present the understanding behind these algorithms and experimental evaluations to compare accuracy measures, speed/accuracy trade-offs, and training methods. The review evaluates the main convolutional neural networks for object recognition.

Contributions and Limitations: Cannot identify object type or size.

Limitation: Needs to be Effective to identify obstacles within 4 meters.

Paper 3: Fusion of Sensor and Vision Systems for Autonomous Navigation

Abstract: It further discusses the advantages and disadvantages of each object sensor model and makes important conclusions. We offer simple visual illustrations summarizing the emergence of styles of object discovery in the under deep literacy. Finally, we identify where future research will be done.

Achievements and Limitations: High cost and difficult hardware set up.

Objectives: To improved depth estimation and object localization.

SCOPE OF THE PROJECT

The limitations of the Neon Nexus assignment are set to reap a completely incorporated, useful, and intelligent attention prototype, specializing in the following middle deliverables:

- **Multi-Sensor Synergy:** The first understanding is to fuse the radar (Arduino) to the digital camera (AI vision) such that it can reliably detect objects. By fusing the sensor modalities, the device is able to not only "see" what an object is but also provide an accurate (distance) reading of the object.
- **Environmental Visualization:** The deployment of the device will visualize objects on a screen radar, the deployed screen will be displayed initially in a range of 8 meters. Thus, the user will have an adequate graphical representation of the present environment.
- **Feedback Grounded Human:** We focus on providing audio and voice feedback to verify device detections. This layer was intelligent: it uses the text-to-speech method to simultaneously announce object names, but announce the distance only when the reading has remained stable for a sufficiently long (3 seconds) period to ignore audible feedback acknowledging a distance due to some random occurrence.
- **Simulating and Testing Real-World Components:** The project verifies the device by simulating and testing the application with physical hardware components. This process includes establishing and persisting continuous connections with the Arduino device using the Web Serial API.
- **User Experience and Interface:** The main component of the work, is building a user experience to interface with your actions.

Solution

The solution is the Neon Nexus v7.4, which acts as a unified, multi-modal sensing platform for enhanced situational awareness. It breaks down the barrier of fragmented normal sensing paradigms through a robust AI integration with physical radar hardware. The Nexus is a "fifth sense" that converts multi-layered sensing and provides structured voice alerts to human action, all within a single highly usable web interface.

This solution is comprised of three fully inclusive middle modules:

1. The AI Vision Module: Object Identification (The "What")

This module produces the context necessary for safe navigation. Rather than actually informing something, it informs what that something is.

Technology Used: The machine makes use of the TensorFlow.Js framework jogging the pre-skilled COCO-SSD version directly within the net browser**.**

Functionality: It processes a live video stream from a standard webcam to perform real-time object and animal recognition. Detected items are instantly classified (e.g., "laptop," "chair," "dog") and displayed on the canvas with bounding boxes and confidence scores.

2. The Hardware Module: Arduino Radar (The "How Far")

While the AI provides identification, the bodily radar affords the critical distance dimension, that's often greater correct than camera-based ranging.

Technology Used: A simple Arduino microcontroller is hooked up to an ultrasonic distance sensor (just like the HC-SR04).

Functionality: The sensor continuously measures the gap to the closest obstacle in centimeters. This information is streamed immediately to the web utility in real-time using the Web Serial API, which acts as the information bridge between the bodily global and the virtual interface.

3. **The Intelligent Voice Feedback Module:**

A Human-Centric Logic.This is what provides meaning to the context of information that a person will be comfortable hearing amid information overload - "beep-fatigue."

Distance Stability Test: To prevent a rapid and conflicting announcement, a governed 3 second stability check was added to the system. The device will only announce the distance (e.g., "Current distance 150cm") if that reading remains within a small tolerance band for the entire three seconds time window.

Immediate Object Context: In contrast, if the AI detects a novel object, the interval towards the name of the object (e.g., "cup detected") will be announced very rapidly (after a short cooldown) so that the user can receive a meaningful piece of high-level information about the object as quickly as possible.

Proximity Warning: In addition, the second varying-rate beeping output will provide non-verbal, haptic feedback, with the interval of beeps occurring closer and closer together, the closer the object moves. Or put another way, the output will be non-verbal in addition to being an immediate visceral warning.

OBJECTIVES

Concentrate on hardware improvement

The goal of the project is to develop a systematic knowledge related to how software and physical sources interact.

Create a Radar Scanning Unit: Build a radar scanner with Arduino and the HC-SR04 ultrasonic sensor with any logical needed to map and store the scanning data.

Write a Web Serial Connection: Build a robust and non-stop conversation hyperlink between the Arduino hardware and the internet application the usage of the Web Serial API to obtain actual-time distance measurements.

1.) Concentrate on AI Integration: The primary purpose here is getting the device to understand (now not simply measure) its environment.

2.) Enable actual object detection in actual time: Implement the software program common sense to allow real object detection with the aid of merging a not unusual digital camera feed to the TensorFlow COCO-SSD model and successfully run inside the browser to allow low latency to the person**.**

3.)Fusing Data: Create the requisite rules to effect the combination of item quality (from the AI) with space assessing (from the Arduino) to provide a coherent piece of information (e.g., "chair at 150 cm").

4.)User Feedback & Intelligence Centered: This is where the human-centred design of the system comes into play to ensure the information provided is useful and not overwhelming.

5.) Voice and beep commentary: The system should contribute voice and beep commentary depending on type and distance. The system should provide voice and beep commentary efficiently.

6.) Clarity of the auditory: Further, the system should employ distance stability check logic, so the distance only speaks when it has been stable for 3 seconds. This is different than if it simply announced the object name which would provide some context quickly but could also lead to auditory clutter.

METHODOLOGY

Developing the Neon Nexus v7.4 follows an organized and circular Methodology based on Design Thinking Principles to ensure that the end product is technically sound, solves real problems for the user. The flow is focused on repeating improvements to the object detection and feedback loop.

1. Identify the needs of the customer and users through Design Thinking Methods:

The first stage concentrates on Empathy. We learned of the main needs for a user needing better spatial awareness: need for context (to know what the obstacle is) and need for calm, reliable distance feedback (no constant loud updates). This stream led us to fuse vision (AI) and ranging (radar) with smart voice logic.

2. Turning the needs of customer into Project / Product:

The need is linked with the Project's architecture (object detection cycle flow)

Input Phase: Needs reliable data streams from both the Camera (to identify object type) and Ultrasonic Sensor data (for distance).

Processing Phase: Requires fast and efficient pre-processing and loading of the TensorFlow COCO-SSD model in the browser environment.

3.)Specification of the Project / Product:

This phase solidified the technical choices based on performance requirements.

Fusion Requirement: Specification to use Web Serial API to connect the physical Arduino sensor data with the front-end JavaScript logic.

Feedback Specification: Definition of the voice feedback logic, specifically the 3-second distance stability check and the immediate object announcement to manage auditory overload.

Visualization Specification: Designing the futuristic radar visualization to clearly map the sensor distance for the user.

4.)Designing the solution for the customer needs:

This involves the physical implementation and testing within the cyclic flow

Detection Phase: The core loop where the AI performs Object Detection and Distance Estimation (using the sensor data when available).

Output Phase (Data Fusion): This critical step performs Sensor-Camera Data Fusion. It is here that the system decides which information to speak and when, according to the stability criteria.

REQUIREMENTS

1.)Hardware Requirements:

- Aurdino UNO
- HC-SOR4 Sensor
- USB-Webcam
- Wires
- Laptop (or) embedded system

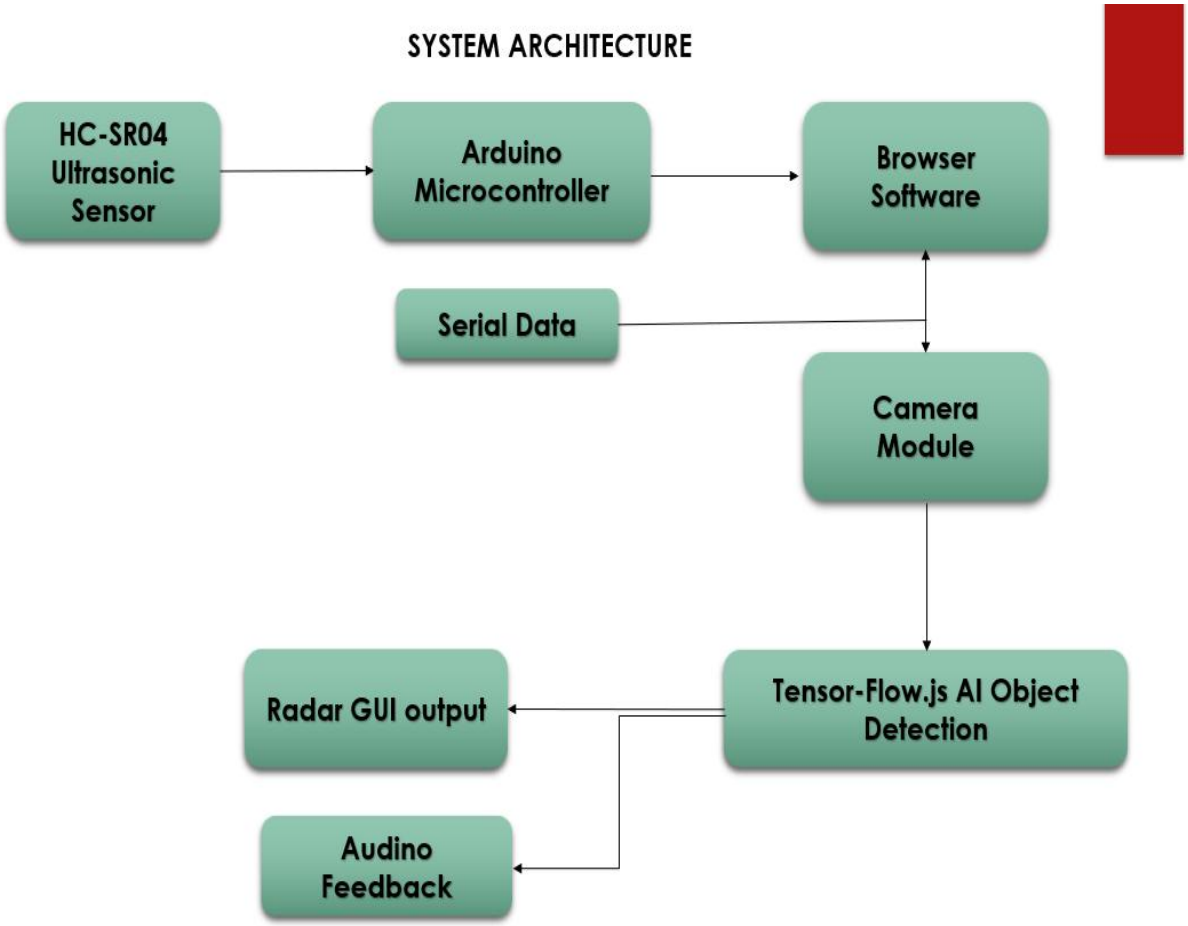
2.) Software Requirements:

- Web Browser
- Arduino IDE
- COCO-SSD Model

3.) Technologies and Tools:

- TensorFlow.js
- Windows
- Git hub
- VS code

DESIGN DIAGRAM



1.)primary Phase: Input devices like Arduino-UNO Ultra-Sonic Sensor and Camera are capture the data and transfer the captured data to tensor-Flow.js AI Object Detection Module.

2.) secondary Phase: Processing units like Browser software and Tensor-Flow.js AI Object Detection(COCO) process or analyze the data which are obtained from input devices. By using existing data and give final required output throughout output devices like Speaker Radar GUI

3.)Final Phase: In final Phase the processed data come from output devices like Speaker and Radar GUI

PROPOSED MODULES

1.)**UI & Control Module (The Cockpit)**

This module manages the entire user-facing interface and handles all system-level interactions.

Function: It is responsible for the overall visual presentation (the neon theme), dashboard metrics (Uptime, Confidence), and the control panel.

Key Tasks: Manages the state of all connection buttons (Connect Arduino, Start Camera, Mute), displays real-time status messages, and initiates the system's core functions based on user input.

2.)Object Detection Module (The AI Eye)

This module handles all elements of pc vision, turning uncooked digital camera feed into contextual records.

Technology: Relies at the TensorFlow.Js library to load and run the COCO-SSD model.

Function: It manages the digital camera movement acquisition (setupCamera), executes the continuous detectObjects loop, and performs immediately visible obligations, which include drawing bounding containers and labels round detected objects at the canvas.

Output: Generates the object's elegance (e.G., "chair") and its prediction confidence rating.**3.)Sensor Integration Module (The Data Bridge)**

This module is the communication link between the physical world and the software application, specifically managing the radar.

Function: It establishes and maintains the serial connection with the Arduino microcontroller via the Web Serial API.

Key Tasks: Executes the robust readSensorLoop to constantly parse incoming data, filter non-numeric characters, and convert the raw serial stream into a clean, digital sensorDistance value (in centimeters).

Control: This will allow the system to run in what we will call Test Mode and simulate sensor data if there is no real data from the hardware.

4.) Intelligent Voice Feedback Module (the Decision-Maker):

This is the heart of the system's innovation, as it is the function that filters, prioritizes, and then communicates the information auditorily to the user.

Function: The updateSensorData function will run continuously, acting as the main method to enable and sustain the state of speech output for what it should communicate**.**

Distance Stability: The distance will be stable checked for only 3 seconds, STABILITY_DURATION, and within a very small tolerance, the distance readings must be stable for 3 seconds before the Web Speech API will announce, to avoid an overload of auditory output due to the temporarily fluctuating distance readings.

NOVELTY

Neon Nexus v7.4 provides a rethinked experience, as it correctly engages through processing technology and shows the smart processing of human centered information through qualifiers of modern lattice-based single sensor paradigms, Designing to be more beneficial to novel aware models. The real benefit is in cross-modal data synthesis and use: the ability to bind high level semantic reasoning with Artificial Intelligence (A.I) vision, and with sub-centimeter spatial accuracy from tightly coupled ultrasonic radar is new developments. The result is higher quality designs of models that are context aware which are certainly improved upon options that are exclusively either vision or exclusively distance options. Another intriguing aspect of the architecture is its useful embodiment of Edge Computing. The significant processing capabilities of a COCO-SSD (and other deep learning applications) is fully realized on the customer-side hardware enable by TensorFlow. This means a customer is not limited by or dependent upon Internet cloud based infrastructures, which ensures community latency is zero, while the attention machine can provide real-time, and therefore highly reliable information, information to inquiring minds in an un-connected low a-head environment. The only new the human centric part is the Intelligent Multi-Modal Outputs of Voice Alerts, proximity.

CONCLUSION

The project Neon Nexus v7.4 was successful in demonstrating the possibility of a true smart climate awareness system by providing edge computing capabilities alongside solid physical sensing elements. Although the original unifying focus was on faster and modular real-time AI-based object recognition and detection capabilities, the project ultimately provided a viable proof-of-concept for various assistive robotics, security, or related technologies based on the speed and low-latency of the system developed.. The main unifying goal of achieving real-time AI-based object recognition with distance measurements from an Arduino-based radar of zero-point-centimeter accuracy was accomplished and embodied within an accessible CDC web app. The most notable feature of the system is the Intelligent Voice Feedback Module. The 3-second distance stability check allowed the project to solve the dilemma of auditory overload by indicating contextually relevant information (the name of object) with stable, accurate spatial information to elevate previous connected sensors to a fully human-centered model for interaction. In summary, Neon Nexus v7.4 not only achieved every objective initially outlined—from building a radar scanner and allowing real-time detection to modular development—but also delivered a quality, low-latency, proof-of-concept for assistive technology to robotics to security potential applications.

Future Scope:

The modularity of the task presents an excellent framework for further refinement including:

1. Enabling SLAM (Simultaneous Localization and Mapping) capabilities for spatial memory and route planning.

2. Facilitating deployment on more powerful dedicated Edge AI platforms (e.g. , Raspberry Pi 5) for better portability and performance.

3. Enhancing the sensing to include LiDAR for sensing for better distance accuracy and robustness to the environment.

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